Cupcakes E-Commerce Website Using Django

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Submitted to:

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1. Introduction

With the rapid growth of the internet and digital technologies, web-based applications have become an essential part of modern businesses. Small businesses such as bakeries and dessert shops are increasingly shifting toward online platforms to promote their products and interact with customers efficiently. This project focuses on the development of a Cupcakes E-Commerce Website using the Django web framework.

The main objective of this project is to design and implement a dynamic website where users can view cupcake products, read customer reviews, and contact the business directly through an online form. The project demonstrates the practical use of Django's Model–View–Template (MVT) architecture, database management using Django ORM, and frontend integration using HTML and CSS.

This system eliminates the need for manual record keeping and provides a structured and scalable solution for managing product data, customer feedback, and user inquiries.

2. Problem Statement

Traditional bakery businesses mostly rely on physical stores or social media platforms such as Facebook and Instagram. While these platforms help in marketing, they do not provide structured data storage, efficient product management, or proper customer interaction mechanisms. Managing customer queries manually is time-consuming and error-prone.

The problem addressed by this project is the lack of a centralized web-based system for managing cupcake products, customer reviews, and contact messages. This project aims to solve these issues by providing a Django-based application that ensures organized data handling, better user experience, and easy scalability.

3. Project Objectives

The primary objectives of this project are:

- To design a dynamic and responsive cupcakes website using Django
- To implement database models for products, reviews, and contact messages
- To use Django ORM for secure and efficient database interaction
- To display dynamic data using Django templates
- To provide a contact form for user communication
- To follow best practices of web development and security

4. Scope of the Project

The scope of this project is limited to displaying cupcake products, showing customer reviews, and receiving user messages. The system does not include online payments or user authentication in the current version. However, the design allows easy expansion in the future.

The project is suitable for:

- Academic purposes
- Small bakery businesses

Learning Django web development fundamentals

5. Technologies Used

5.1 Python

Python is used as the main programming language due to its simplicity, readability, and extensive library support.

5.2 Django Framework

Django is a high-level Python web framework that encourages rapid development and clean design. It provides built-in security features, ORM, and template engine.

5.3 Frontend Technologies

HTML for webpage structure

CSS for styling and responsiveness

JavaScript for basic interactivity

5.4 Database (SQLite)

SQLite is used as the database during development. It is lightweight, fast, and suitable for small to medium-scale applications.

6. System Architecture (MVT Pattern)

The project follows Django's Model–View–Template (MVT) architecture:

6.1 Model

Models define the structure of the database. Each model represents a table, and each field represents a column.

6.2 View

Views handle business logic. They retrieve data from models, process user requests, and send responses to templates.

6.3 Template

Templates manage the user interface. They display dynamic content using Django Template Language (DTL).

This architecture ensures separation of concerns, making the system easy to maintain and extend.

7. Database Design and Models

7.1 Product Model

The Product model stores information about cupcakes available on the website.
Fields include:

- Name of the cupcake
- Description
- Original price
- Discount percentage
- Product image
- Creation date

Methods are implemented to calculate discounted prices dynamically.

7.2 Review Model

The Review model stores customer feedback.
Fields include:

- Customer name
- Review message
- Rating (1 to 5 stars)
- Optional image
- Date of review

This model helps build customer trust and engagement.

7.3 ContactMessage Model

This model stores user inquiries submitted through the contact form. Fields include:

- Name
- Email
- Phone number
- Message
- Timestamp

8. Views and Business Logic

The home view is the core of the application. It performs multiple tasks:

- Fetches latest products from the database
- Fetches customer reviews
- Handles GET and POST requests
- Saves contact messages using Django ORM
- Displays success messages using Django Messages Framework

The view ensures smooth communication between the backend and frontend.

9. Frontend Design and Templates

The frontend is designed with a clean and user-friendly layout. The main template (index.html) displays products, reviews, and the contact form.

Key features:

- Dynamic product rendering using {% for %} loops

- Conditional display of discounted prices

- Responsive design using CSS

- Form handling with CSRF protection

10. Security Considerations

Security is an important aspect of this project. Django provides built-in protection against common web vulnerabilities.

Implemented security measures include:

CSRF protection for all forms

ORM-based database queries to prevent SQL injection

Server-side validation of user input

11. Testing Strategy

The application was tested locally using Django's development server. Manual testing was performed to verify:

- Product display
- Review rendering
- Contact form submission
- Database entries
- Responsive UI behavior

Testing ensured that the system functions correctly under normal usage.

12. Deployment

Currently, the project is deployed locally using Django's built-in development server. The system can be deployed on cloud platforms such as:

- PythonAnywhere

- Heroku
- AWS

Deployment would allow public access and real-world usage.

13. Challenges Faced

Several challenges were encountered during development:

- Understanding Django's MVT architecture
- Managing static and media files
- Connecting backend logic with frontend templates
- Handling form submissions securely

These challenges helped improve practical understanding of Django.

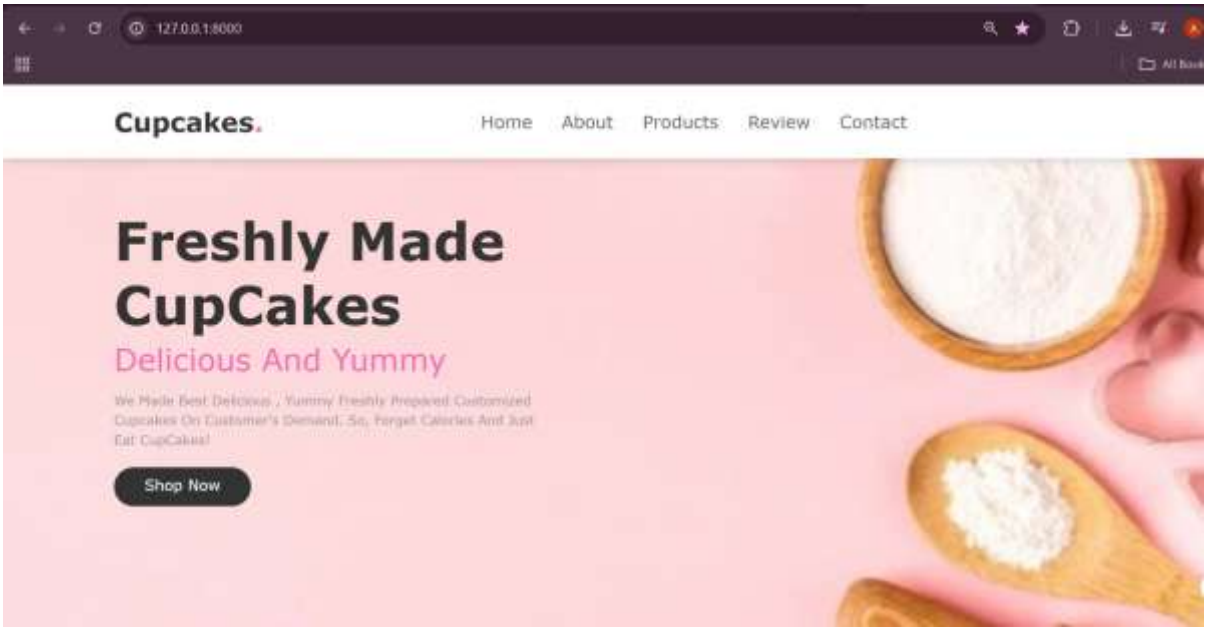
14. Future Enhancements

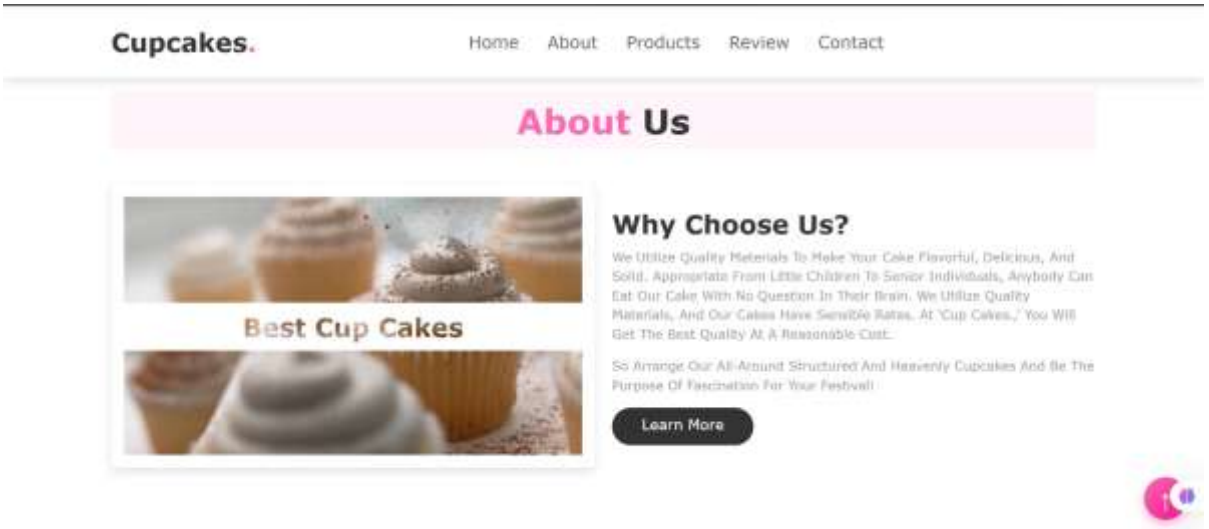
Possible future improvements include:

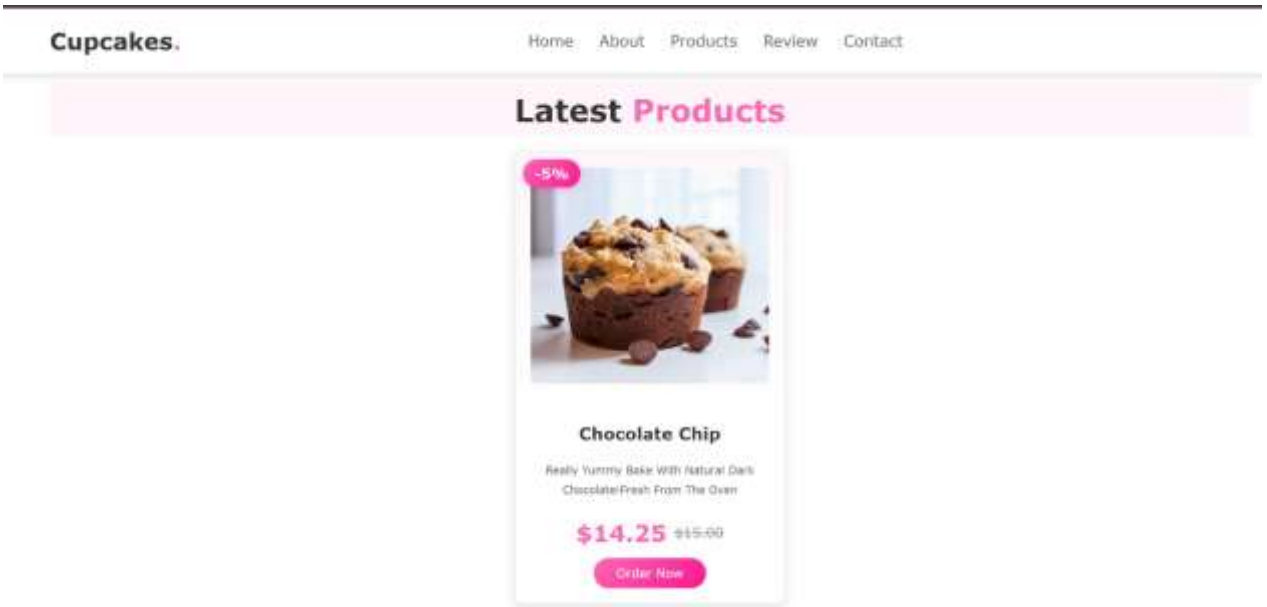
- User registration and login system
- Shopping cart functionality
- Online payment integration
- Admin dashboard customization
- Mobile application version

15. Conclusion

This project successfully demonstrates the development of a Django-based Cupcakes E-Commerce Website. It fulfills academic requirements and showcases practical skills in web development, database management, and frontend-backend integration. The system is scalable and can be enhanced further for real-world use.






Cupcakes.
Home
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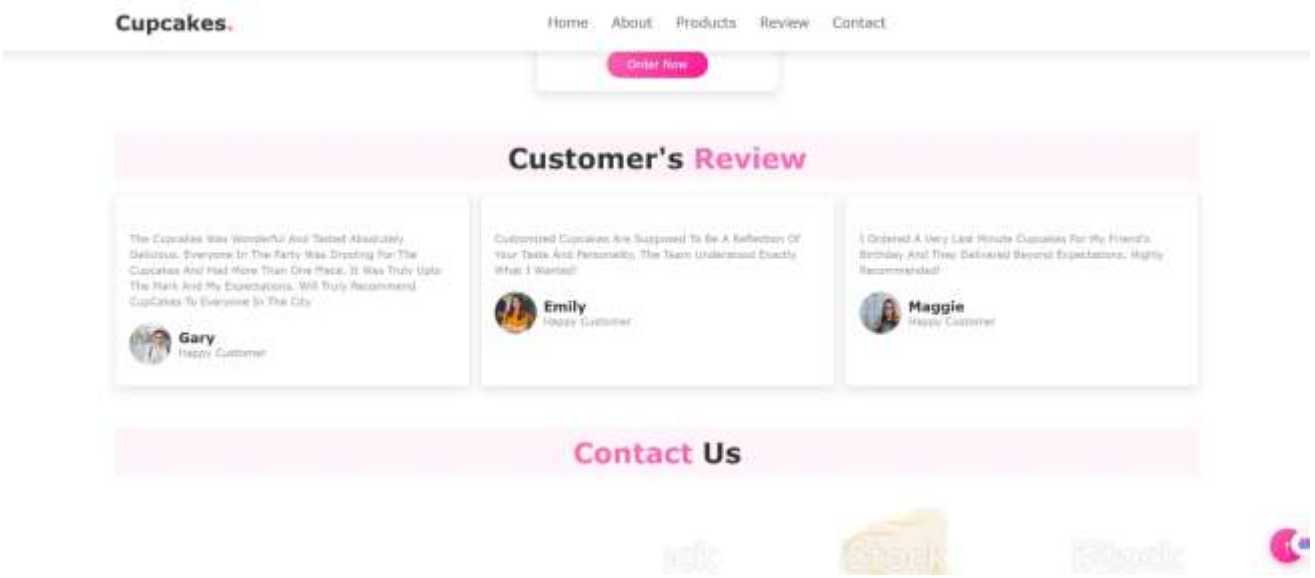
Chocolate Chip
Really Yummy Bake With Natural Dark Chocolate Fresh From The Oven
$14.25 $15.00
Order Now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Gary
Happy Customer
Emily
Happy Customer
Maggie
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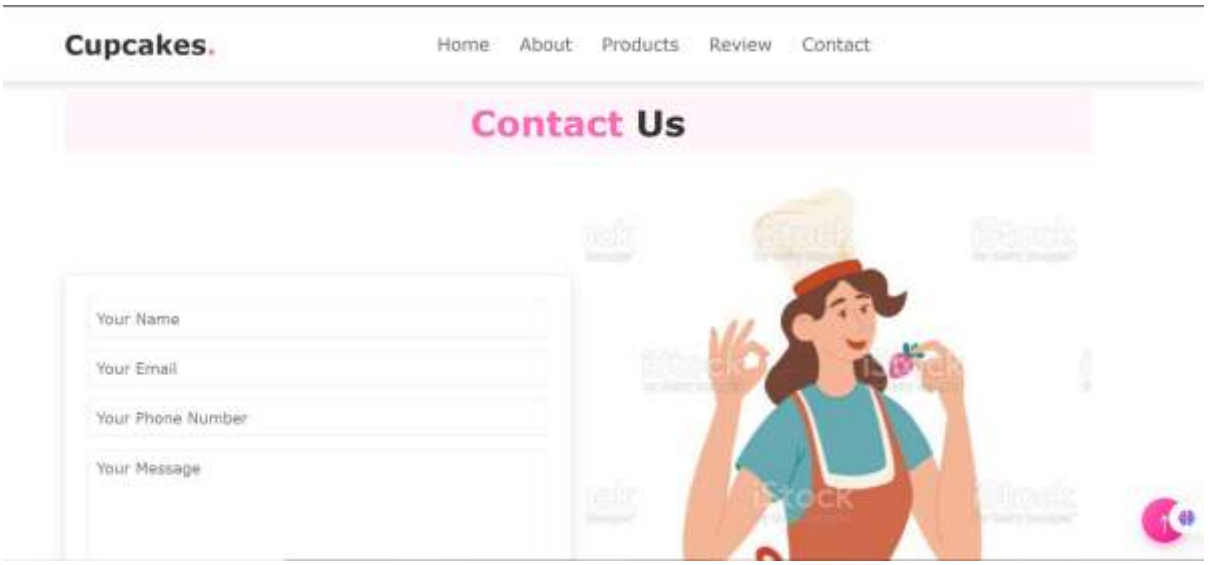
Contact Us


Cupcakes.
Home
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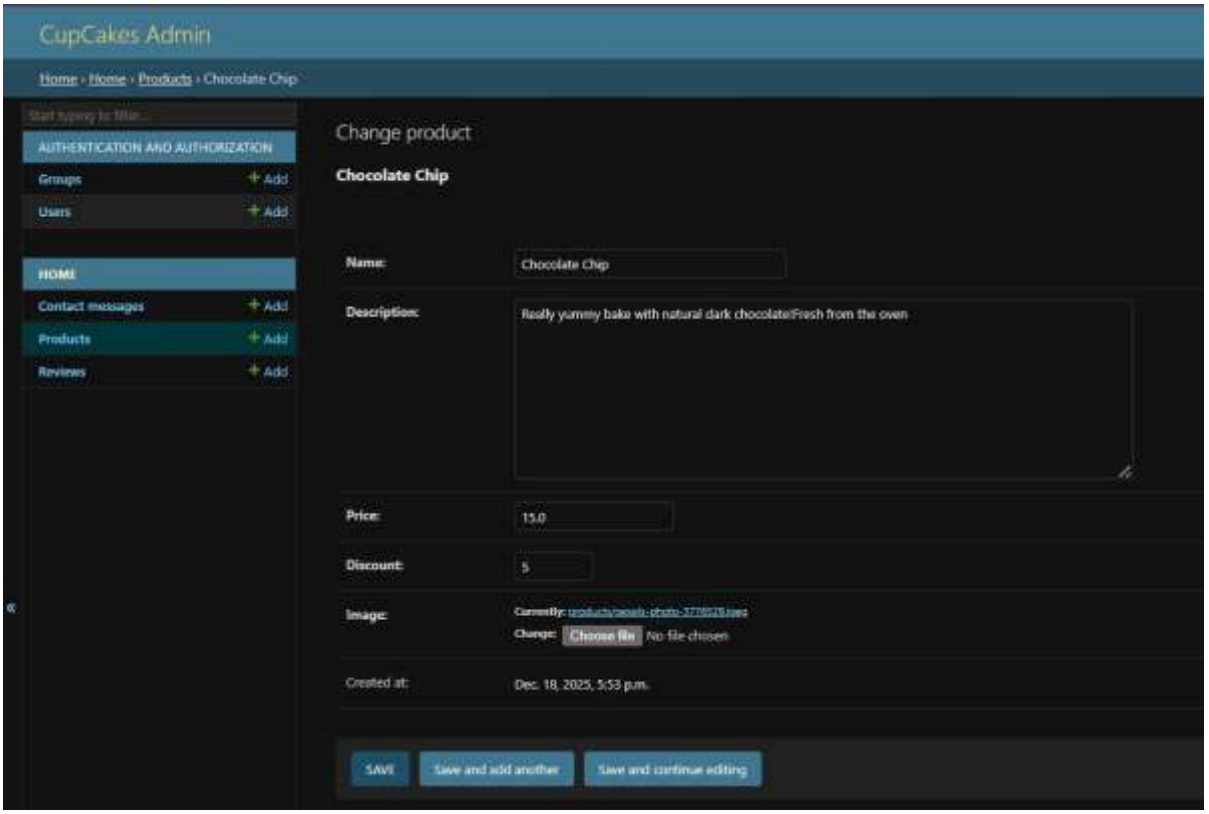
CupCakes Admin
Home › Home › Products › Chocolate Chip
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
HOME
Contact messages + Add
Products + Add
Reviews + Add
Change product
Chocolate Chip
Name:
Chocolate Chip
Description:
Really yummy bake with natural dark chocolate!Fresh from the oven
Price:
15.0
Discount:
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SAVE
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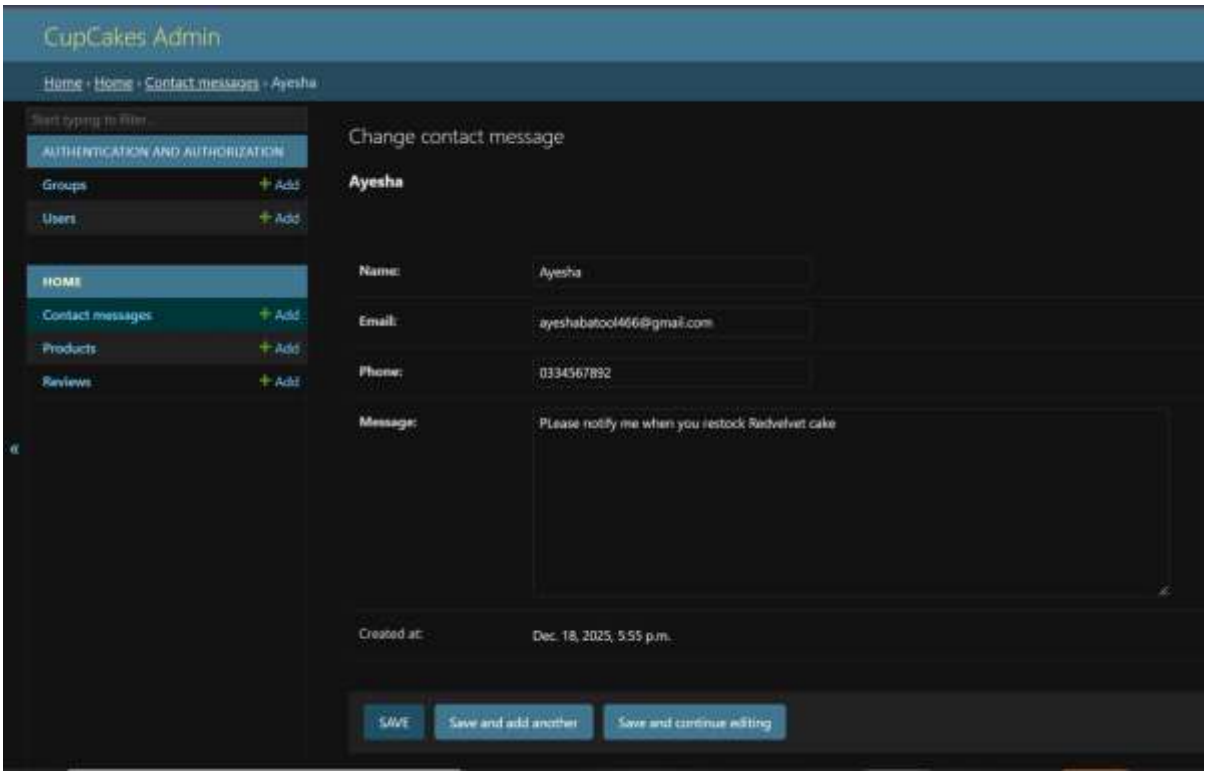
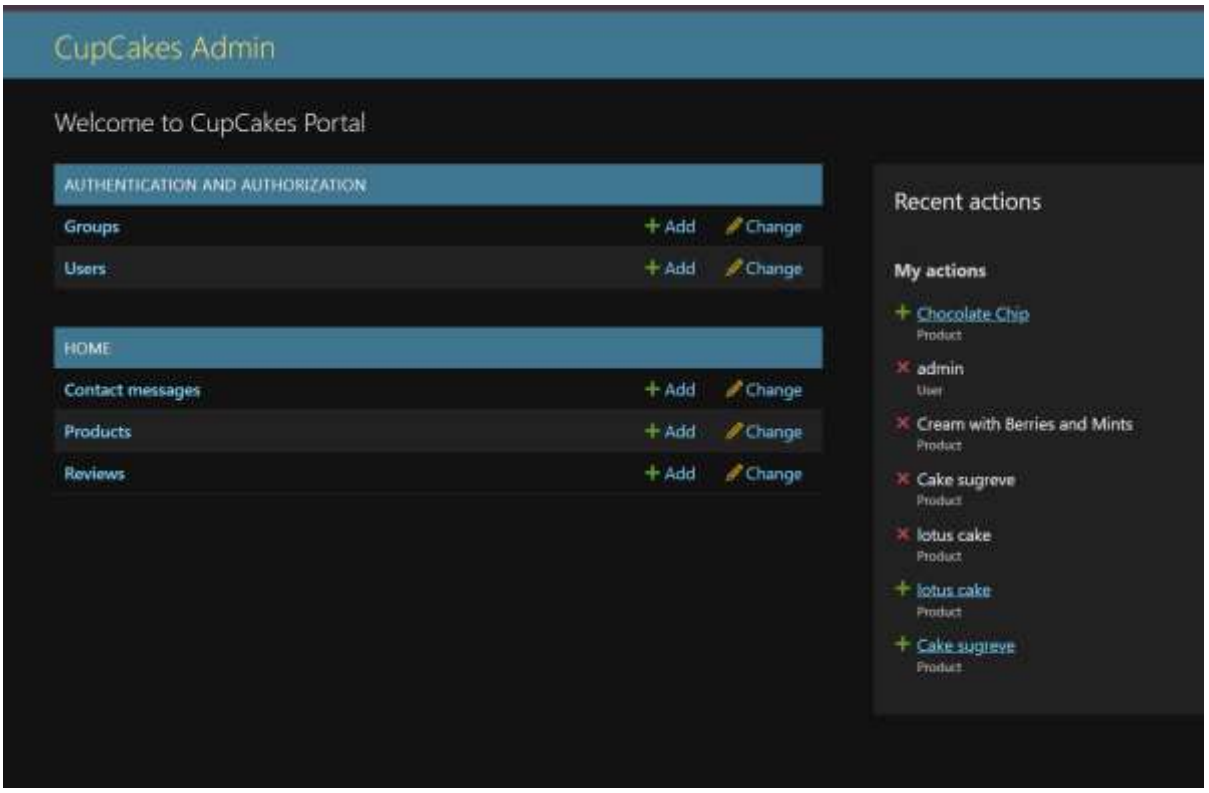

CupCakes Admin
Home › Home › Contact messages › Ayesha
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
HOME
Contact messages + Add
Products + Add
Reviews + Add
Change contact message
Ayesha
Name: Ayesha
Email: ayeshabatool466@gmail.com
Phone: 0334567892
Message: PLease notify me when you restock Redvelvet cake
Created at: Dec. 18, 2025, 5:55 p.m.
SAVE
Save and add another
Save and continue editing


CupCakes Admin
Welcome to CupCakes Portal
AUTHENTICATION AND AUTHORIZATION
Groups + Add Change
Users + Add Change
HOME
Contact messages + Add Change
Products + Add Change
Reviews + Add Change
Recent actions
My actions
Chocolate Chip
Product
admin
User
Cream with Berries and Mints
Product
Cake sugreve
Product
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Product